# VISHAKH CHERUPARAMBATH

Looking for a position in the field of AI which demands the best of my personal ability in terms of technical and analytical skills to contribute to the growth of organization.

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Address - **Bäumenstr. 4, 90762 Fürth**
Born on 27.06.1997 in Kerala, India

## LANGUAGES

1. English - Fluent C1
2. German - Intermediate B1

## EDUCATION

**1. Masters in Electromobility-ACES** *(2022 - present)*
*Friedrich-Alexander-Universität Erlangen*
*Major: Artificial Intelligence*
*Key elements: Machine Learning, Pattern Recognition, Data Science, Autonomous systems*

**2. Bachelors in Mechanical Engineering** *(2015 - 2019)*
*Amal Jyothi College of Engineering, Kerala, India*

## DIGITAL SKILLS

C++, Python, HTML, CSS, JavaScript, Bash

Power Automate, SQL, Snowflake, GIT, GitHub, Gitlab, Confluence, AI Agents, MS Office

Azure AI Foundry tools, Llama Index, Gpt 4o, LaTeX, Linux, ROS 2, Rviz, YOLO v7, Faster RCNN, PyTorch, Pandas, Tensorflow

## ABOUT ME

1. Automotive Enthusiast
2. Travel Blog
3. Music
4. Photography

Fürth, 05 Aug. 25

## WORK EXPERIENCE

### Werkstudent Process Automation and Data Analysis
*(07/2024 – 05/2025)* ***Siemens Mobility GmbH, Erlangen***

- Developed VBA macros to automate repetitive tasks for process efficiency.
- Explored multi-agent LLM architectures from SiemensGPT to build use case bots for configuration management team.
- Created Power BI kanban dashboard for rolling stock project insight using effective semantic models.
- Contributed to SAP GUI scripting, managed databases using SQL, and snowflake to support AI-powered applications.
- Utilized Power Automate for dataflow automation and reporting.
- Build SharePoint pages for AI and Automation internal tool use cases.

### Student Assistant - Vision-Based Interaction for Autonomous Mobile Robots (Projektarbeit)
*(11/2024 – 05/2025)* ***Institute FAPS, Erlangen***

- Developed and implemented algorithms for robotic applications.
- Optimized gesture recognition & behavior tree models.
- Integrated multi-object detection model using ZED 2i stereo camera API.
- Enhanced DevOps workflows with CI pipelines and software stack in FAPS GitLab.
- Created comprehensive user test cases and performed qualitative analysis.

### Apprenticeship - Project Management
*(08/2020 – 09/2021)* ***Hindustan Aeronautics Limited, Bangalore***

- Managed product database of 11 assembly modules using IFS ERP.
- Assigned job ticket and documentation of process workflow for engineering, and MRO.
- Prepared requirements, data pre-processing, and build interactive dashboards for workflow analysis and reporting.

## ACADEMIC PROJECTS

### Road Lane Tracking and Object Detection
Estimated drivable space using output of semantic segmentation neural network with RANSAC plane fitting, detected and filtered lanes and obstacles using OpenCV, and computed minimum distance to impact by fusing depth and 2D object detections.

### Visual Odometry in Autonomous Driving
Implemented visual odometry using Essential Matrix decomposition with SIFT and FLANN, achieving 92% accuracy in camera motion estimation. Reduced outliers by 85% using RANSAC and reconstructed vehicle trajectory via incremental pose transformations.

### dsssGPT - AI Assistant with Telegram Bot
Built a Telegram AI assistant using a local LLM (TinyLlama model) with LangChain-inspired message routing and GPU-accelerated inference. Enabled real-time user interaction and deployed end-to-end using Python. Utilized telegram bot API keys to integrate bot setup.

### Smart Quality Monitoring in Industry Automation
Developed SVM and CNN models for electric motor fault classification, achieving 96% test accuracy. Improved CNN performance by 12% via hyperparameter tuning and data augmentation, with robust preprocessing and deployment.

### Classification of Road Traffic Image Challenge
Performed image-based classification of signal behaviour (stop vs. non-stop) using PyTorch, achieving over 80% accuracy. Integrated object detection, signal classification in complex traffic scenarios. Replaced conventional prediction methods with a robust computer vision-based solution.